

Assignment: Visiting Card Digitization & Voice Notes Orchestrator

Overview

Welcome to the AI Engineer assessment! In this assignment, you will build an end-to-end system for automating the digitization of visiting cards via a Chat UI. The system will extract data, log it into Google Sheets, ensure data uniqueness, handle voice recordings, and send real-time notifications.

You must orchestrate this entire workflow using a **LangGraph agent**. The agent will act as the core brain of the system, calling tools to handle UI interactions, data extraction, database operations, and external communications.

 **Time Expectation:** You have 48 hours to complete this assignment. We do not expect production-ready perfection, but rather a functional, well-architected prototype that demonstrates your understanding of agentic workflows, API integrations, and modern cloud deployment.

Business Scenario

Imagine a sales team that attends conferences and frequently collects physical visiting cards. They need a quick, frictionless way to digitize this information on the go and add custom voice notes about the contacts they meet.

Your system needs to:

1. Provide a Chat UI where users can upload an image of a visiting card.
2. Use a single LangGraph agent to manage the conversation state and orchestrate the tasks below.
3. Extract contact details (Name, Phone, Email, Company, etc.) from the image.
4. Log the extracted data into a designated Google Sheet.
5. Check the Google Sheet to ensure no duplicate entries are created.
6. Send a WhatsApp message via the WhatsApp Business API to a specific manager alerting them that a new card entry has been logged.
7. Allow the user to send a voice recording via the Chat UI to be processed and updated in the Google Sheet for that specific contact entry.

Core Requirements & Tasks

Task 1: Chat UI Setup

Design and implement a Chat interface that accepts user inputs.

- Must support image uploads (for visiting cards).

- Must support audio/voice note uploads.
- Must be able to manage multiple chat sessions

Task 2: Single LangGraph Agent Orchestration

Design a single LangGraph agent that controls the application logic.

- The agent must maintain a conversation state to seamlessly link subsequent voice notes to the previously uploaded visiting card.
- All external integrations (extraction, Google Sheets, WhatsApp) should be implemented as tools or nodes within this graph.

Task 3: AI Data Extraction

Implement an extraction pipeline within the agent to process the uploaded visiting card.

- Extract structured data containing at minimum: Name, Phone Number, Email Address, and Company.

Task 4: Google Sheets Integration & Deduplication

Design a synchronization mechanism to use Google Sheets as your primary database.

- Implement a robust deduplication logic to check if a contact already exists before adding a new row.
- Handle duplicate cases gracefully within the user interface via the agent.

Task 5: Voice Recording Handling

Develop a workflow for handling voice notes sent after a card is uploaded:

- Process the audio data so it is accessible
- The agent must associate the processed audio data with the correct corresponding entry in the Google Sheet.
- The existing entry should be updated with audio URL

Task 6: WhatsApp Notification Integration

Integrate the **WhatsApp Business API** for real-time alerting.

- Trigger a WhatsApp message to a predefined phone number upon the successful logging of a unique card.

Task 7: Cloud Deployment

Deploy your complete backend system to a cloud platform.

- **Containerization:** Ensure the application is properly containerized for deployment.
- **Secrets Management:** Implement secure handling of all API keys, service accounts, and credentials.



Technical Constraints & Stack

You must adhere to the following technological constraints for this assignment:

- **AI Orchestration:** LangGraph Agent
- **Backend API:** FastAPI (Python)
- **Frontend :** React JS
- **Database:** MongoDB (Atlas M0 Free Tier) or Cloud SQL / managed SQL equivalents.
- **Notifications:** WhatsApp Business API
- **Data Storage:** Google Sheets API
- **AI & Extraction:** Candidate's choice of Vision and Audio processing tools.
- **Hosting:** Candidate's choice (e.g., GCP Cloud Run, AWS, Render).



Bonus Points (If you have extra time)

- **Human-in-the-Loop:** Implement an interrupt in the LangGraph agent to ask the user for manual confirmation (e.g., "Are these extracted details correct?") before writing to Google Sheets or sending the WhatsApp notification.
- **Data Enrichment:** Automatically find and append the company's website or LinkedIn profile based on the extracted company name.



Deliverables

1. **Source Code:** A link to a public/private GitHub repository.
2. **Deployed URLs:** Links to the live, working Chat UI and backend.
3. **Documentation (README.md):**
 - Clear instructions on setting up environment variables.
 - Step-by-step instructions to build, run, and test the project locally.
 - Deployment guide and a brief architectural overview of your LangGraph agent setup.
4. **Demo Video (Loom/YouTube):** A 3-5 minute video demonstrating:
 - Uploading a visiting card image to the Chat UI.
 - The agent successfully extracting and logging it to Google Sheets without duplicates.
 - The WhatsApp notification arriving via the WhatsApp Business API.
 - Uploading a voice note and seeing the Google Sheet update with the new information via the agent's context.



Evaluation Criteria

Criteria	Description
Agentic Logic & LangGraph (30%)	Effective use of a single LangGraph agent to orchestrate the entire extraction, deduplication, and notification flow.
System Integration (20%)	Smooth communication between Chat UI, AI services, Google Sheets, and WhatsApp Business API via FastAPI.
Logic & Deduplication (20%)	Robust handling of duplicates and proper state management to link voice notes to the right contact.
Cloud Deployment (20%)	Correct, secure deployment. Emphasis on containerization and secret management.
Code Quality (10%)	Clean, modular code structure, properly documented FastAPI endpoints and graph state.